

## 1 Privacy is a Mitigation — the setup

Data → algorithm → model → predictions serve the world. Privacy is a way to **mitigate** harm from the data. We want to protect individuals while keeping the model **useful**.

**Cryptographic solutions** to privacy: ✗ (too heavy / not the right fit).

**Statistical solutions** to privacy: ✓ — this is the DP approach.

## 2 Randomized Response (Local DP)

Each person reports the **truth with probability  $e^\epsilon/(1+e^\epsilon)$**  and a **falsehood with probability  $1/(1+e^\epsilon)$** . No one needs to be trusted — noise is added **before** the data leaves the individual.

**Privacy guarantee:** for two datasets differing in one person, the ratio of report probabilities is bounded by  $e^\epsilon$ :  $\Pr[\text{RR}(x)=b] / \Pr[\text{RR}(x')=b] \leq e^\epsilon$ .

### Two tensions to balance:

- **Privacy** — hide individual data
- **Utility** — keep aggregate answers accurate

Two trust settings: **Randomized Response** (no trusted curator — local DP) and **Trusted Curator** model (central DP).

### Utility:

- De-bias the noisy reports to estimate the true fraction
- $E[\text{estimate}] = \text{the true mean (unbiased)}$
- Deviation from truth is  $O(1 / \sqrt{n})$  with high probability

**Neighbouring datasets**  $X$  and  $X'$  **differ by only one element** — this is what the privacy proof relies on (Q10).

**Assignment:** min truthful-answer probability in RR (from the slide's algorithm) = **50%** (Q4). Neighbouring datasets **differ by only one element** (Q10). To maximise **utility**:  $\epsilon$  minimum for privacy,  $\epsilon$  **maximum** for utility (Q5).

## 3 The Trusted-Curator Model & $\epsilon$ -DP

**Trusted curator** holds raw data, adds noise centrally, then releases. This is **Differential Privacy** proper (a Cynthia Dwork idea).

$\epsilon$ -**DP definition:** a mechanism  $M$  is  $\epsilon$ -differentially private if, for **all neighbouring datasets**  $X, X'$  (differ in one row) and all output sets  $S$ :

$$\Pr[M(X) \in S] / \Pr[M(X') \in S] \leq e^\epsilon$$

Smaller  $\epsilon$  = stronger privacy, less utility. This ratio bound is **Cynthia Dwork's definition** (Q9).

**Trust scenario (Q7):** in the DP curator model you **Trust the curator, Do NOT trust the world** — the curator sees raw data; the released output must protect against everyone else.

**Why a trained model can leak (Q2):** even if raw data isn't released, **an adversary can reverse-engineer sensitive training data** from the model (membership/attribute inference).

**Why cryptographic solutions don't work for model privacy (Q1):** it's unclear whether all complex functions can be effectively evaluated in the encryption space, and evaluation in encrypted space is computationally expensive.

## 4 The Laplace Mechanism

For a numeric function  $f: X \rightarrow \mathbb{R}$ , release  $f(X) + \text{noise}$  where noise  $\sim \text{Laplace}(0, \Delta/\epsilon)$ .

**Sensitivity**  $\Delta$  = the **maximum absolute difference**  $|f(X) - f(X')|$  over all neighbouring datasets  $X, X'$  (Q8). It captures how much one person can change the output.

**Laplace density:**  $f_{\text{Lap}}(x) = (1/2b) e^{-|x|/b}$ , with scale  $b = \Delta/\epsilon$  (the privacy parameter). Bigger  $\Delta$  or smaller  $\epsilon \rightarrow$  more noise.

### Key relationships (exam):

- **Higher sensitivity (large  $\Delta$ )  $\rightarrow$  more noise added** (Q11)
- Neighbouring sets differ by one element — same as before
- Laplace utility (error):  **$O(1/\epsilon n)$**

**Randomized Response vs Laplace utility:** RR error is  $O(1/\epsilon \sqrt{n})$ ;

Laplace error is  $O(1/\epsilon n)$ . So to reach the **same** privacy-utility trade-off, the **Laplace mechanism needs quadratically FEWER people** (Q12) — the trusted-curator model is more efficient.

**Assignment quick answers:** sensitivity = **max absolute difference of  $f$  over neighbouring datasets** (Q8); DP ratio bound =  $e^\epsilon$  (Q9); high sensitivity  $\rightarrow$  **more noise** (Q11); Laplace needs **quadratically fewer people** than RR (Q12).

## 5 Worked Example (Q6)

100 people surveyed on how they voted; Randomized Response with  $\epsilon \approx 0.5$ . Of 100 respondents, 60 reported 'Yes' and 40 reported 'No'.

Estimate the true number who actually voted 'Yes', rounded to the nearest whole number  $\rightarrow$  **answer  $\approx 51$** . (Use the RR de-biasing / unbiased-estimator formula: recover the true count from the noisy counts using the report probabilities, then round.)

## 6 One-line recap for the exam

- **RR (local DP):** add noise before data leaves you; no trusted party; utility error  $O(1/\epsilon \sqrt{n})$
- **Trusted curator (central DP):** curator adds noise; more efficient
- $\epsilon$ -**DP:**  $\Pr[M(X) \in S] / \Pr[M(X') \in S] \leq e^\epsilon$
- **Neighbouring datasets differ by one element**
- **Sensitivity**  $\Delta = \max |f(X) - f(X')|$  over neighbours
- **Laplace mechanism:** add  $\text{Laplace}(0, \Delta/\epsilon)$  noise; error  $O(1/\epsilon n)$
- Higher sensitivity  $\rightarrow$  more noise; smaller  $\epsilon \rightarrow$  more noise
- Trust: **trust curator, not the world**